Toward Human–Inspired Collaborative Intelligence for Human–AI Team Decision-Making: A Survey

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As AI systems increasingly participate in complex decision-making, effective human–AI (HAI) collaboration requires more than accurate recommendations or calibrated reliance. Humans and AI systems must coordinate goals, communicate uncertainty, share control, and adapt over repeated interactions. Yet, existing HAI decision-making research often studies these mechanisms separately, limiting our understanding of HAI as a dynamic team process. This survey examines human–AI joint decision-making through the lens of human-inspired collaborative intelligence. Drawing on human team decision-making, cognitive science, HCI, robotics, and AI, we synthesize work on mutual modeling, communication and alignment, control allocation, and team-level evaluation. We introduce a task taxonomy organized by increasing coordination demand and show that dynamic team tasks remain underdeveloped in current HAI research. Based on this synthesis, we propose a Human-AI Joint Decision-Making framework as a literature-grounded conceptual framework that characterizes HAI decision-making as an adaptive cycle linking mutual modeling, bidirectional communication, control sharing, and team-utility-based evaluation. We conclude with open challenges for building robust, adaptive, and accountable human–AI teams.

1 Introduction

As AI systems become increasingly embedded into everyday decision-making, they are no longer used only as passive tools for prediction or recommendation. In many domains, AI systems now participate in ongoing decision processes in which humans and AI systems exchange information, revise beliefs, allocate responsibility, and jointly determine actions. Human–AI (HAI) decision-making, therefore, refers to settings in which humans and AI systems jointly contribute to forming, evaluating, and executing decisions through ongoing interaction and mutual influence [44, 84].

However, much of the existing work in HAI decision-making continues to frame AI primarily as an assistive or supervisory technology where the AI recommends, explains, predicts, or optimizes, while the human accepts, rejects, monitors, or overrides [20, 53, 76, 84, 136]. This framing has produced important gains in task performance and decision accuracy [54, 84, 152], but it captures only part of what makes collaboration effective. In sustained decision-making contexts, success depends not only on whether an AI system is accurate but also on whether human and AI team members can coordinate their goals, communicate uncertainty, build appropriate trust, adapt to one another's behavior, and negotiate control as task demands change.

These challenges are not new. Decades of research on human-human team decision-making have examined how teams coordinate under uncertainty, distribute cognitive work, maintain shared mental models, adapt roles, and recover from misalignment [22, 80, 101, 124, 158]. Foundational work in human teamwork has shown that **effective teams are dynamic systems characterized by mutual understanding, role negotiation, communication, and continuous adaptation** [10, 19, 50, 62, 85, 123, 132, 146]. This literature offers an important but underutilized foundation for HAI.

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Rather than treating collaboration as a collection of isolated mechanisms, such as explanation, trust calibration, or task allocation, it suggests that effective **HAI decision-making should be understood as an integrated team process**.

Recent work has increasingly recognized that HAI decision-making is jointly produced rather than simply the result of human oversight or AI recommendation. Cassady [26] emphasizes alignment through shared context, objectives, and constraints grounded in common mental models, while As' ad et al. [3] proposes a healthcare-focused framework for explainable, reasoning-based joint HAI decision-making in which AI acts as a facilitator. Similarly, McGrath et al. [103] models trust as a bidirectional process shaped by communication during collaboration, and Shen et al. [127] synthesizes approximately 400 studies to redefine alignment as mutual HAI adaptation. Punzi et al. [117] similarly identifies the need for frameworks that account for HAI co-evolution [114]. While these studies establish the importance of shared context, trust, bidirectional alignment, human modeling, and communication for joint decision-making, existing surveys and frameworks often examine these mechanisms separately, focus on domain-specific forms of collaboration, or emphasize individual components of the HAI relationship, such as trust, explainability, or alignment. As a result, the literature still lacks a cohesive account of how these mechanisms interact over time within dynamic HAI teams. In particular, less attention has been paid to how human-inspired principles of team decision-making can inform HAI systems that must coordinate, communicate, adapt roles, and share control across sequential or evolving decision contexts.

This survey addresses these gaps by organizing the current landscape of HAI joint decision-making through the lens of human-inspired collaborative intelligence. We first review HAI approaches that model human characteristics, support human–AI communication and alignment, and allocate control or decision authority between human and AI team members. We then examine how HAI team performance has been evaluated, arguing that accuracy alone is insufficient for assessing adaptive joint decision-making. Building on this synthesis, we introduce a task taxonomy that organizes HAI decision-making settings by increasing coordination demand and use it to motivate the HAI-JDM framework introduced in Section 6. Finally, we identify open challenges and research directions for designing more cohesive, adaptive, and team-oriented HAI decision-making systems.

The **contributions** of this survey are threefold. First, we synthesize research on human team decision-making and HAI collaboration to **clarify how principles of human teaming can inform the design of HAI joint decision-making system**s. Second, we organize the **surveyed literature** around core mechanisms of HAI teaming, human and agent modeling, communication and alignment, control allocation, and team-level evaluation, **and identify recurring gaps** across these areas. Finally, we **introduce a task taxonomy and the HAI-JDM framework** as literature-grounded tools for understanding HAI joint decision-making as an adaptive, interactive, and team-based process.

We organize the surveyed literature around four human-inspired mechanisms of HAI joint decision-making: human and agent modeling (Section 2), communication and alignment (Section 3), control allocation and decision authority (Section 4), and team-level evaluation (Section 5). These dimensions guide the structure of the survey and allow us to compare how existing work operationalizes different aspects of HAI teaming. After reviewing the literature through these dimensions, we synthesize recurring patterns, limitations, and dependencies into a task taxonomy and the HAI-JDM framework introduced in Section 6. Lastly, we discuss open challenges in Section 7 and conclude in Section 8.

1.1 Methodology

We followed a narrative and integrative review methodology to synthesize interdisciplinary work on human-human team decision-making and human-AI joint decision-making (HAI-JDM). We define AI systems as systems with some degree of autonomy in perception, reasoning, prediction, adaptation, or action selection. Because HAI-JDM spans HCI, AI, robotics, cognitive psychology, decision science, and organizational behavior, our goal was not to conduct a narrow

systematic review of a single domain, but to identify recurring mechanisms, assumptions, and evaluation practices relevant to human-inspired collaborative intelligence.

We searched Google Scholar, Semantic Scholar, Allen AI, ACM Digital Library, IEEE Xplore, arXiv, and major AI/HCI venues including CHI, CSCW, IUI, HRI, NeurIPS, ICML, ICLR, AAAI, and IJCAI. Search terms included combinations of "human-AI decision-making," "joint human-AI decision-making," "human-AI collaboration," "human-AI teaming," "shared control," "human modeling," "theory of mind," "bounded rationality," "trust calibration," "communication," "alignment," "reinforcement learning from human feedback," and "team decision-making." We began with survey papers, framework papers, and highly cited empirical studies, then expanded the corpus through backward and forward citation tracing. The initial search was conducted in November 2025, with periodic follow-up searches through publication to capture recent work, including emerging work on agentic AI.

We included papers that contributed theoretical, computational, empirical, or evaluative insights into HAI joint decision-making, human team decision-making, or mechanisms relevant to HAI teaming. Papers relevant to the following themes were included: human or agent modeling, communication and alignment, shared control or decision authority, team adaptation, or evaluation of HAI team performance. We prioritized peer-reviewed work but included highly relevant preprints that addressed emerging areas not yet well represented. We excluded work that did not involve meaningful HAI interaction or joint decision-making, including Wizard-of-Oz systems and purely scripted robotic behaviors. Selected work from adjacent decision-support domains was included when its findings generalized to HAI-JDM, particularly for trust, reliance, feedback, and control allocation. Our final corpus contains 173 papers spanning human team decision-making, HAI decision-making, alignment and communication, shared control and leadership, evaluation, emerging agentic AI systems, and survey or framework papers.

2 Leveraging Human Characteristics for More Natural HAI Models

Effective HAI joint decision-making requires AI systems to reason not only about the task but also about the humans with whom they collaborate. In human teams, successful collaboration depends on understanding teammates' goals, beliefs, capabilities, limitations, and changing states over time. Similarly, HAI systems that treat human behavior as noise, static input, or perfectly rational optimization risk mischaracterizing the collaborator they are designed to support. This section reviews computational methods that integrate existing theories and understanding of human characteristics and behaviors into HAI models to facilitate more natural collaboration. We focus on three dimensions that are central to human-inspired teaming: mental models and Theory of Mind (Section 2.1), bounded rationality (Section 2.2), and trust (Section 2.3). Collective discussion of these dimensions shows that humans should be modeled as adaptive decision-makers whose beliefs, preferences, cognitive constraints, and expectations evolve through interaction. Accounting for these characteristics is a prerequisite for HAI systems that can coordinate with humans as teammates rather than function only as assistive tools.

2.1 The Effect of Mental Models on HAI Collaboration

Collaborative decision-making requires team members to reason about both the task and one another. When individuals collaborate, they form beliefs about their teammates' goals, knowledge, intentions, capabilities, and reliability. These beliefs are often described as **mental models**, which are internal representations of external reality that support reasoning, prediction, and action [48]. First introduced by Kenneth Craik in 1943 [35], mental models have become central to understanding human reasoning, prediction, and decision-making [72]. In team settings, mental models include representations of collaborators. **Theory of Mind (ToM)** refers to the ability to infer others' beliefs, intentions,

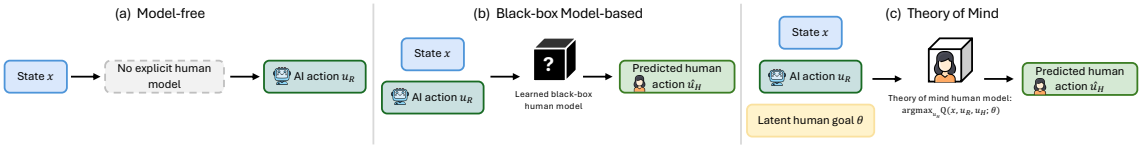


Fig. 1. Adapted from Choudhury et al. [31] to illustrate three increasingly structured approaches to human modeling in HAI. Model-free approaches map the current state directly to an AI action without explicitly modeling the human collaborator. Black-box model-based approaches predict human actions from states and AI actions using a learned human model. Theory of Mind approaches further represent a latent human goal, predicting human action by modeling the human as a goal-directed decision-maker.

and knowledge states in order to predict and interpret their behavior [21]. First formalized by Premack and Woodruff [116], ToM allows individuals to interpret social cues, reason about shared knowledge, and anticipate teammates' decisions. As shown in Figure 1, ToM differs from more general behavioral modeling because it represents collaborators as intentional agents rather than as black-box sources of action. In collaborative contexts, this enables teammates to update expectations, coordinate under uncertainty, and move toward shared decisions [55]. This process of mutual modeling extends directly to HAI collaboration. **Effective AI teammates must model not only what humans do, but also why they act, what they believe, and how their beliefs change through interaction.** At the same time, humans form evolving mental models of AI capabilities, limitations, and reliability. This section reviews work on ToM and mental models in HAI, emphasizing how mutual modeling can support more adaptive and coordinated HAI decision-making.

2.1.1 AI Agents Trained Using Theory of Mind Principles Perform Better in Collaborative Tasks. A central limitation of many collaborative AI systems is that they are trained with partners that do not resemble humans. Agents trained through self-play, for example, learn by interacting with copies of themselves and may therefore develop conventions, strategies, or assumptions that fail to transfer to human collaborators [139, 165]. Even population-play methods, which train agents with multiple artificial partners, may not adequately capture the variability, bounded rationality, and social conventions that characterize human behavior. As a result, agents that perform well with artificial partners can still coordinate poorly with humans. Strouse et al. [139] address this limitation by proposing Fictitious Co-Play, a training method that exposes agents to diverse partner strategies and skill levels rather than relying only on self-play or population play. As shown in Figure 2, Fictitious Co-Play trains agents to adapt to a broad pool of independently trained partner policies, improving collaboration with both simulated human-like agents and real human partners. In an Overcooked-style task, the method outperforms behavioral cloning using completed deliveries as a team-performance metric, while also considering human partner preference and behavioral coordination [139]. These findings suggest that preparing AI systems for human collaboration requires accounting for variation in human strategies, conventions, and skills. Other work similarly demonstrates the value of incorporating human behavior into collaborative agent training. Carroll et al. [25] show that agents trained using human behavioral data outperform self-play agents when paired with human partners and can even improve HAI team performance beyond human–human baselines, evaluated primarily through team score in an Overcooked-style coordination task. More recent work has explored scalable approaches to modeling human reasoning, including automated Bayesian inverse planning methods that infer human goals, beliefs, and actions more robustly, typically evaluated through goal-inference accuracy, action-prediction accuracy, or likelihood of observed human behavior [170]. Yu et al. [163] models human behavior using ToM within a multi-agent Markov Decision Process framework related to Cooperative Inverse Reinforcement Learning (CIRL) [56] (see 2.2.1). In human-subject studies, their ToM-based models outperform perfectly rational and level-k human-reasoning baselines,

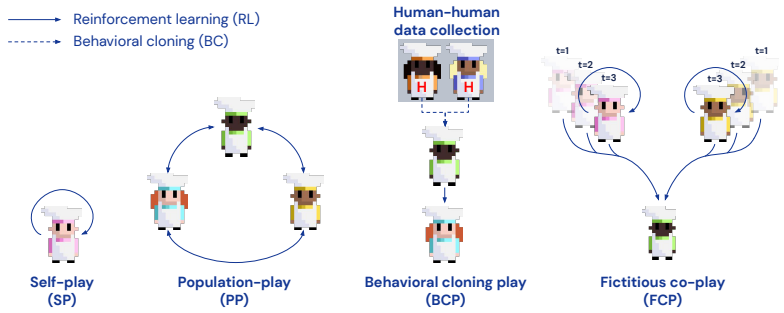


Fig. 2. Training paradigms for collaborative agents, adapted from Strouse et al. [139], with permission. Self-play trains agents with copies of themselves, population play trains agents with a population of artificial partners, behavioral cloning play uses human data to construct partner models, and Fictitious Co-Play trains agents against diverse self-play partners and checkpoints to improve adaptation to human collaborators.

evaluated through team performance, human-action prediction, collaborative reward, and participants' ability to infer AI actions. The results suggest that modeling human beliefs and intentions can help AI systems anticipate, explain, and coordinate with human partners over time. Overall, these studies show that human modeling is not only an interpretability tool, but a practical mechanism for improving coordination in HAI decision-making.

2.1.2 Humans Prefer Collaboration With AI Agents Trained Using Theory of Mind. ToM-based modeling shows particular benefits in high-coordination tasks, where successful collaboration depends on anticipating a partner's intentions and adapting accordingly. Across several settings, humans report stronger preferences for collaborating with agents that incorporate ToM-like reasoning, independent of objective task outcomes [134, 154, 166]. These studies evaluate ToM benefits using both task-level outcomes and human-centered measures, such as team performance, perceived understanding, perceived service quality, perceived humanness, and user preference or willingness to collaborate. In real-time collaborative environments, Zhang et al. [167] introduce DPT-Agent, which combines finite-state machines, code-as-policy reasoning, and ToM-based reflection to support HAI teamwork. Evaluated in the Overcooked environment, the agent outperforms rule-based and language-agent baselines on task-performance measures such as game score, efficiency, and latency, while also improving human-centered evaluations of cooperation and preference [27].

2.1.3 Humans Mental Models of AI Affect Collaboration. A growing body of HAI research examines how humans form, update, and rely on mental models of AI systems during interaction. This perspective shifts attention away from treating humans as passive users or fully rational decision-makers and instead asks how people understand the AI systems with which they collaborate. Berretta et al. [15], Kelly et al. [78], Kim and Cho [79] introduce latent-variable approaches to measure human beliefs about AI capability, showing that humans often overestimate AI performance, perceive AI competence as more uniform than it actually is, and update their beliefs slowly even after receiving feedback. Studies show automation bias and the machine heuristic can lead users to overvalue AI recommendations or assume machine outputs are objective and reliable [94, 107, 133, 142]. Conversely, algorithmic aversion can lead users to discount AI advice, especially after observing errors [37, 60]. These studies typically evaluate mental models through belief-estimation tasks, calibration between perceived and actual AI capability, changes in beliefs after feedback, and downstream reliance or decision behavior. These findings suggest that human collaborators may enter HAI interactions

with persistent, sometimes inaccurate assumptions about AI capabilities, which can shape reliance, coordination, and decision quality.

Complementary theoretical work examines how human mental models of AI develop through repeated interaction [59, 65, 141]. This work identifies several components of AI mental models, including users' understanding of the task domain, awareness of AI reasoning processes, and beliefs about when human and AI capabilities are complementary. These perspectives help explain why common interventions, such as explanations, confidence scores, or trust-calibration mechanisms, can have inconsistent effects when they do not address the broader mental models through which users interpret AI behavior. These studies show that human beliefs about AI are dynamic, consequential, and often imperfect. Their evaluation approaches also illustrate that mental models should be assessed not only through subjective understanding or trust, but also through whether humans can accurately predict AI behavior, calibrate reliance, and update beliefs over time. **Designing effective HAI systems, therefore, requires accounting not only for how AI systems model humans, but also for how humans model AI systems over time.**

Overall, this literature shows that ToM and human modeling can improve HAI collaboration by enabling AI systems to anticipate human actions, adapt to diverse partners, and support more interpretable coordination. At the same time, these benefits depend on the quality, scalability, and robustness of the underlying human models. Future HAI systems will need methods that capture human variability without overfitting to narrow assumptions about rationality, expertise, or preference structure.

2.2 Using Human Bounded Rationality Rather than Assuming Optimality

Many formal models of decision-making begin from the assumption that decision-makers are rational agents who select actions that maximize expected utility given their preferences, available information, and beliefs about the environment [51, 161]. This assumption also underlies many models of artificial agents, which are often defined in terms of rational action or reward maximization [122]. While useful for formal analysis, this view is limited when applied to human team decision-making. Human collaborators rarely have complete information, unlimited cognitive resources, or stable access to all possible alternatives and consequences. As a result, models that assume fully rational human behavior can mischaracterize how people actually make decisions, coordinate with others, and respond to uncertainty [131].

Human decision-making is instead shaped by **bounded rationality**, wherein people make decisions under constraints of limited information, limited time, uncertainty, and finite cognitive capacity. Rather than computing globally optimal actions, humans often rely on heuristics, approximations, and context-dependent strategies that are efficient but imperfect [36]. In team settings, these constraints are amplified because individuals must reason not only about the task, but also about teammates' knowledge, intentions, reliability, and likely actions. Human teams are therefore not collections of perfectly rational agents, but adaptive systems whose decisions emerge from incomplete information, cognitive shortcuts, communication patterns, and negotiated understandings of the task.

2.2.1 Modeling Bounded Rationality Improves Sequential HAI Decision-Making. Bounded rationality is especially relevant in HAI settings that require sequential collaboration, where humans and AI systems must repeatedly act, observe one another, and update their strategies over time. These problems are commonly modeled using reinforcement learning (RL), in which agents learn decision policies through interaction with an environment or expert and reward-based feedback [75]. In Inverse Reinforcement Learning (IRL), an agent observes demonstrations from an assumed expert and infers the reward function that explains those demonstrations [110]. Although IRL provides a foundation for learning from human behavior, it treats the demonstrator as an optimal actor, which is a poor fit for human collaborators

whose decisions reflect bounded rationality, incomplete information, and cognitive constraints. In collaborative HAI settings, RL models that explicitly incorporate bounded-rationality assumptions can better capture human behavior and improve joint task performance compared with models that assume fully optimal human action [43]. Similar assumptions also appear in shared-control paradigms, which we discuss in Section 4.

Hadfield-Menell et al. [56] demonstrated the utility of bounded rationality in Cooperative Inverse Reinforcement Learning (CIRL), which formalizes HAI alignment as a two-player game between a human H and a robot R (AI agent) that share a common reward. Unlike standard IRL, CIRL incentivizes the robot to help the human achieve the human's objective and incentivizes the human to act pedagogically so the robot can infer that objective. Formally, CIRL extends a Markov decision process M to include a hidden reward parameter θ, known to the human but not initially known to the robot: $M = (X, A_H, A_R, T, \Theta, r_\theta, P_0, \gamma)$, where X is the state space, A_H and A_R are the human and robot action spaces, T is the transition function, Θ is the set of possible reward parameters, r_θ is the reward function, P_0 is the initial state distribution, and γ is the discount factor. In this setting, the human and robot repeatedly observe the state, choose actions, and receive a shared reward. Their goal is to coordinate their policies so that the team achieves high cumulative reward over time. Because θ is hidden from the robot, the robot maintains and updates a belief over possible human objectives as it observes human actions. This formulation captures an important feature of collaboration, which is that human actions can be both task-directed and communicative. For example, in a cooking task such as Overcooked or ChefWorld [43, 140], a human may prepare an ingredient not only to make progress on the task, but also to reveal to the robot which meal they intend to make.

However, CIRL still relies on idealized assumptions about human behavior. Pedagogic Value Alignment extends this framework by modeling humans as noisily rational rather than perfectly optimal [43]. In this approach, human actions are modeled with a Boltzmann-rational policy, where higher-value actions are more likely but not guaranteed: $\pi_H^{\odot}(a_H|x, b_R, a_R; \theta) \propto \exp(\beta Q(x, b_R, a_H, a_R; \theta))$. Here, b_R represents the robot's belief over the reward parameter, Q estimates the value of joint human–robot actions, and β controls the assumed degree of human rationality. Lower values of β represent noisier or less consistently optimal human behavior. The robot then updates its belief over θ based on how likely the observed human action is under this noisily rational policy, and selects actions that maximize expected value under its current belief. This formulation preserves the cooperative structure of CIRL while better reflecting the fact that human collaborators may communicate imperfectly, act approximately, or rely on partial information. It is evaluated through collaborative task success in ChefWorld, where incorporating bounded rationality substantially improves performance. The simulated human-robot team achieves a 97% success rate compared with 46% under IRL [43]. This result illustrates the broader evaluation point that HAI systems may perform well under rational-agent assumptions yet fail under more realistic human behavior models. Bounded-rationality modeling is therefore important not only as a cognitive theory, but also as a practical design principle for improving team-level performance in sequential decision-making.

2.3 Trust: The Foundation of Collaboration

Trust is central to collaboration because it shapes whether team members are willing to rely on one another under uncertainty [18, 150]. Trust shapes how individuals rely on teammates and incorporate their contributions during joint decision-making. Interpersonal trust is often defined as the expectation that another individual will act in a way that supports a shared goal, despite vulnerability and uncertainty [102]. From this perspective, trust is not simply a stable attitude toward another collaborator, but a mental representation of that collaborator's reliability, competence, intentions, and likely future behavior [45]. These representations influence whether individuals accept teammates'

input, defer to their judgment, or seek additional information before acting. Trust judgments also draw on Theory of Mind processes, as individuals infer others' intentions and assess whether their actions are likely to align with team objectives [42, 121]. Trust is consequential for HAI decision-making because both over-trust and under-trust can degrade collaboration. Over-trust may lead humans to accept flawed AI actions, while under-trust may prevent them from benefiting from useful AI support [2, 52, 83, 96, 97].

2.3.1 Trust in AI Is Shaped by Early Performance and Interaction History. In human teams, first impressions can strongly influence later assessments of trustworthiness [45]. Similarly, empirical studies of HAI collaboration show that trust in AI develops through interaction and is strongly shaped by early experiences with system behavior. Tolmeijer et al. [148] examine trust in decision tasks where participants could choose whether to follow AI recommendations. Their results show that early system performance has a persistent effect on later trust. Initial incorrect recommendations significantly reduce trust, while later perceived improvements can only partially restore it. They also find that most dispositional traits have limited influence on trust, with the exception of age and affinity for technology. These findings suggest that early reliability is especially important in HAI collaboration because initial failures can create durable mental models that limit later reliance. Other work shows that trust in AI depends not only on accuracy, but also on how the AI system presents itself and participates in collaboration. Humans are more willing to trust and collaborate with AI teammates when they are competent and transparent about their identity, which can improve both task performance and human perception of the interaction [164]. In some cases, humans may even report greater willingness to trust an AI teammate than a human one [38], potentially reflecting automation bias, the machine heuristic, or different expectations for human and AI collaborators. These findings reinforce that trust is not simply a response to objective performance, but it is shaped by expectations, perceived competence, transparency, and prior beliefs about AI.

2.3.2 Trust Models Evolve Over Time. In both human–human and human–AI teams, trust develops through repeated interaction, feedback, and observed performance. However, humans do not enter HAI interactions with the same expectations they bring to human teammates. Initial trust in AI systems may differ in level, basis, and stability from interpersonal trust [38, 49]. Initial trust may emerge from cognitive heuristics that reduce social complexity and decision burden [11]. Over time, however, collaborators refine their trust assessments as they observe behavior, receive feedback, and compare new evidence against prior expectations [147]. In collaborative decision-making, this dynamic updating is critical. While [1, 70] show trust evolves dynamically over longer interactions, Li et al. [90] argues that purely rational models fail to capture human trust dynamics, which are shaped by both cognitive and emotional processes. They propose a hidden Markov model that learns trust evolution from human acceptance and rejection of AI recommendations. In a study with 245 participants performing income-prediction tasks, their model more accurately captured trust dynamics than established baselines. However, the approach assumes immediate feedback and is evaluated in a general-knowledge setting, leaving open questions about whether similar models generalize to domain-specific, high-stakes, or delayed-feedback HAI environments.

Trusting behavior can occur independently of cooperation [102], but sustained coordination requires both appropriate reliance and shared understanding. Trust, therefore, serves as the relational counterpart to cognitive mental models, linking beliefs about a teammate's intentions and competence to concrete decisions about reliance, communication, and coordination [10, 135]. Overall, these studies show that **trust in AI is experience-driven, path-dependent, and dynamically updated over time**. For HAI joint decision-making, this means that trust should be modeled and evaluated not only through static self-report ratings, but also through longitudinal measures such as reliance behavior, acceptance or rejection of AI recommendations, recovery after errors, and changes in human beliefs about AI capability.

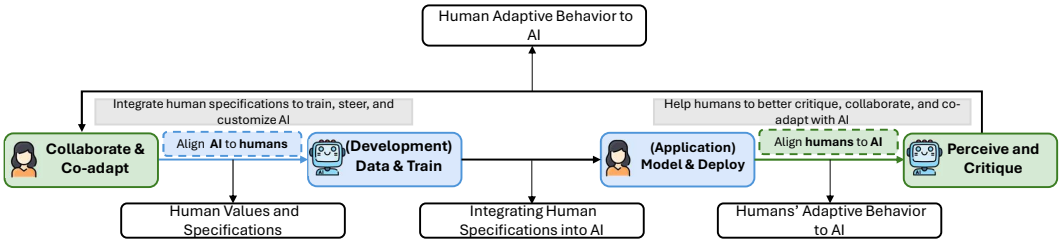


Fig. 3. Bidirectional human-AI alignment framework from Shen et al. [126]. Their framework emphasizes that alignment is not only a process of adapting AI systems to human values and goals, but also of supporting humans in understanding, critiquing, and adapting to AI systems over time.

3 Communication and Alignment as Joint Optimization

Communication is central to team decision-making because it allows collaborators to exchange information, resolve uncertainty, coordinate strategies, and repair misalignment. In human–human teams, communication supports the development of shared goals and shared mental models. In HAI teams, it plays a similar role by enabling humans and AI systems to update their beliefs about one another and about the task. HAI joint decision-making requires ongoing coordination of beliefs, goals, and actions. In this section, we treat communication as the primary mechanism through which coordination is achieved, and bidirectional alignment as the resulting objective. Following Shen et al. [126], bidirectional alignment refers to the interconnected process of aligning AI systems to humans while also helping humans understand, critique, and adapt to AI systems. This alignment also shapes trust dynamics during collaboration [103] and mutual HAI adaptation [127].

Communication is especially important because collaborative decision-making does not automatically imply aligned goals or motivations. Early theories of group decision-making often assumed that groups were primarily motivated by cooperation, but empirical work on mixed-motive decision-making challenges this assumption. Toma and Butera [149] show that competitive motives within groups can reduce the exchange of unshared information and make individuals less willing to revise existing beliefs, even in tasks that require information sharing for successful group performance. These findings suggest that coordination failures often arise not only from insufficient information, but also from misaligned incentives, trust, and interpretations of the task. This insight is directly relevant to HAI decision-making. HAI teams may also experience misalignment between human goals, AI objectives, institutional incentives, and task constraints. Even when both human and AI components are designed to contribute to the same decision, failures in communication can lead to inappropriate reliance, hidden disagreement, poorly calibrated trust, or conflicting interpretations of the objective. Effective HAI collaboration, therefore, requires communication mechanisms that do more than transmit recommendations or explanations; they must support the ongoing alignment of goals, beliefs, uncertainty, and decision authority. Figure 3 illustrates this broader view of bidirectional alignment, in which both humans and AI systems adapt through interaction.

3.1 Implementing Coordination Through Communication Methods

3.1.1 Communication Requirements Increase with Task Interdependence. Communication has long been recognized as a core mechanism of group decision-making. In Leavitt [85] early empirical work, communication structure was shown to influence group accuracy, activity, satisfaction, and organization. Subsequent research further demonstrates

that communication is especially important for complex decision-making tasks, where group members must exchange information, evaluate alternatives, and build consensus [62]. For relatively simple tasks, communication may involve little more than stating, explaining, or defending individual decisions. In more complex tasks, however, communication allows team members to pool distributed knowledge, surface unshared information, and construct a more complete understanding of the decision space [63].

This relationship between communication and task complexity has direct implications for HAI teams. Communication should not be treated as a fixed design feature, but as a coordination mechanism whose importance depends on the interdependence of the task. In pooled tasks, where human and AI contributions can be combined with limited interaction, minimal communication may be sufficient. In reciprocal or highly interdependent tasks, where each teammate's actions shape the other's decisions, bidirectional information sharing becomes essential. HAI systems designed for joint decision-making therefore need communication mechanisms that support not only recommendation delivery, but also clarification, uncertainty exchange, feedback, and shared strategy formation.

3.1.2 Implicit and Explicit Communication Shape HAI Coordination. A growing body of work highlights the importance of communication in human–AI collaboration, particularly the role of implicit signals that humans naturally use during teamwork. In human teams, collaborators often coordinate through cues that are not explicitly stated as instructions, such as gaze, motion, timing, facial expressions, indirect language, or contextual hints. These cues allow teammates to infer intentions, anticipate actions, and adjust behavior without interrupting the task. AI systems often struggle in socially complex settings because they may fail to interpret or generate these subtle communicative signals [89, 91, 162]. As humans increasingly treat AI collaborators as social partners, these implicit scripts of interaction become more important for effective coordination [159].

Several HAI systems attempt to incorporate implicit communication into collaborative decision-making. Liang et al. [91], for example, introduce a model that uses linguistic deduction to exchange implicit strategic hints in the cooperative game Hanabi. Although the model achieved performance comparable to traditional entropy-based approaches, human collaborators were significantly more likely to believe they were interacting with another human. Other work explores implicit communication through action and motion, such as robot navigation behaviors that signal intent or guide human responses without explicit verbal instruction [28, 81]. These approaches suggest that implicit communication can support coordination by allowing humans and AI systems to infer intent from behavior, rather than relying only on explicit commands or explanations. However, implicit communication also introduces evaluation challenges. Some studies measure success through task performance or coordination efficiency, while others rely on human-centered outcomes such as perceived humanness, partner preference, or perceived collaboration quality.

Explicit communication has also been studied through explanations and rationale generation, which allow AI systems to communicate decision-relevant information more directly. Bansal et al. [8] evaluate saliency maps, feature-based explanations, and confidence scores in a human-subject classification task. Although explanations improved human understanding and individual accuracy, they did not consistently improve team performance. In some cases, explanations increased human acceptance of incorrect AI recommendations, reducing joint performance despite unchanged AI accuracy. Similarly, Ehsan et al. [39] show that post-hoc rationales can increase human confidence and perceived understanding of an AI system's strategy, particularly when the rationale communicates long-term planning. However, because such rationales may not reflect the system's true reasoning, they can improve perceived understanding without necessarily improving calibrated reliance or team outcomes. Explanation studies therefore often evaluate communication

through perceived understanding, explanation usefulness, trust or confidence, reliance behavior, and downstream team performance.

These findings suggest that communication is necessary but not sufficient for effective HAI collaboration. Implicit signals, explanations, rationales, and confidence information can all support coordination, but their value depends on whether they help humans form accurate mental models, calibrate reliance, and adapt their behavior appropriately. The variation in evaluation highlights a recurring challenge for HAI communication research: **communication can improve perceived understanding or confidence without necessarily improving coordination, alignment, or team-level performance**.

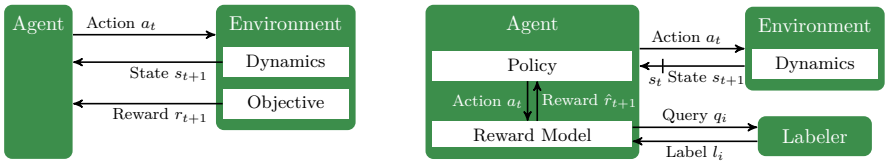
3.2 Implementing Human-AI Team Goal Alignment Through Feedback

3.2.1 Feedback Supports Team Learning and Alignment. Feedback plays an important role in decision-making because it allows individuals and teams to update beliefs, correct errors, and adjust future strategies. However, its effects differ across individual and group contexts. In individual decision-making, positive feedback can produce inconsistent learning effects [40], while negative feedback may support learning but does not necessarily lead to optimal adjustment [82]. These findings suggest that feedback is not automatically useful, but rather its value depends on how it is interpreted and incorporated into future decisions.

In team decision-making, feedback can be especially valuable because it supports shared learning and coordination. Empirical studies comparing individuals and teams show that teams often benefit most from feedback after each decision, whereas individuals may perform better with partial feedback [5, 57, 64, 146]. Increasing feedback frequency can also reduce individual confidence, but appears to have less negative effect on teams. This advantage depends on effective intra-team communication. Feedback improves team performance when members can discuss errors, update shared understanding, and reach consensus about how to adapt. For HAI teams, this suggests that feedback should be treated not only as a training signal for AI systems, but also as a communication mechanism that helps humans and AI systems align goals, recalibrate trust, and coordinate future decisions.

3.2.2 Human Feedback Aligns AI-Training Objectives with Human Preferences. Feedback can shape HAI collaboration not only during interaction, but also during model training. A central example is reinforcement learning from human feedback (RLHF), in which human evaluations are used to define or refine reinforcement learning objectives [33, 77, 172]. RLHF is relevant to broader HAI decision-making settings where goals are difficult to specify formally, demonstrations are incomplete, or reward functions risk misrepresenting human intent. In this sense, RLHF treats feedback as a communication channel through which humans convey preferences, goals, and corrective signals to AI systems. The distinction between traditional RL and RLHF is shown in Figure 4. RLHF has recently been popularized in the context of large language models (LLMs) [172].

Early work by Maclin and Shavlik [99] showed that structured human advice can improve RL performance by guiding Q-learning. Although this approach relied on constrained advice formats, it established the value of human input for accelerating policy learning. Subsequent work further shows that human feedback is not merely a neutral reward signal, but is shaped by social expectations and interaction dynamics. Thomaz et al. [145], for example, found that users often provide positive and anticipatory feedback when training RL agents, adjusting their guidance as they learn about the agent's behavior and treating the agent as a collaborative partner. These findings suggest that **collaborative RL systems must account for the timing, asymmetry, and social meaning of human feedback.**



(a) The standard RL setting.

(b) RLHF with reward modeling.

Fig. 4. Traditional reinforcement learning (RL) and reinforcement learning from human feedback (RLHF), from Kaufmann et al. [77], with permission. In RLHF, human judgments are used to learn or refine a reward model, allowing the system to optimize behavior toward human preferences when the true reward function is difficult to specify directly.

Because feedback-based alignment aims to improve both model behavior and agreement with human preferences, RLHF and related approaches are typically evaluated using both model-performance and human-preference metrics. Depending on the setting, these include task reward, policy performance, sample efficiency, preference-prediction accuracy, human preference ratings, and win rates in pairwise comparisons. These metrics reflect an important feature of feedback-based alignment, where success is not only whether the model improves on a benchmark, but also whether its behavior better matches human preferences and supports effective downstream collaboration.

Despite its effectiveness, RLHF has important limitations for joint human–AI decision-making. One limitation is scalability. Feedback quality and availability can constrain performance, especially when expert judgment is expensive or difficult to obtain. A growing line of work addresses this challenge through reinforcement learning from AI feedback (RLAIF), in which a prompted LLM provides reward signals during training [86, 87]. Their direct RLAIF approach achieves performance comparable to RLHF in summarization and dialogue tasks while reducing reliance on costly human labeling, suggesting that AI-generated feedback may offer a scalable proxy for human preference judgments. Although RLAIF appears to replace human feedback with AI feedback, the feedback model is itself shaped by prior human data, instructions, and preferences. Thus, RLAIF does not fully remove humans from the alignment loop, but instead, it introduces an additional layer that approximates human judgment at scale. For HAI decision-making, this raises the question of whether AI-mediated feedback preserves human intent or introduces new distortions as feedback passes through multiple models.

A second limitation is that RLHF can encode the values, assumptions, or vulnerabilities of the feedback process itself. McIntosh et al. [104] show that RLHF-trained language models can remain vulnerable to ideological manipulation through subtle prompting and may reflect the values embedded in their training or oversight processes. This limitation is especially important for HAI decision-making, where the goal is not only to produce preferred outputs but to support robust, accountable, and context-sensitive collaboration.

Beyond RLHF and RLAIF, feedback can also enter HAI systems through more direct forms of expert guidance. Human preferences can be incorporated before or during training through expert-designed rewards, demonstrations, or data-refinement procedures. Barata et al. [9], for example, show that Deep-Q models trained with dermatologist-defined reward tables improve diagnostic accuracy and melanoma detection sensitivity compared with supervised learning models and unaided clinicians. Clinicians supported by the RL system also achieved higher diagnostic accuracy than clinicians without RL support, demonstrating the potential value of embedding expert knowledge directly into training objectives. Related approaches use expert demonstrations or guidance during training in domains such as autonomous driving, video games, and precision robotics [115, 119, 153, 171].

Expert-guided feedback is also increasingly used outside of RL. Bi et al. [16] use expert preferences to guide programs that refine training data before it is used by an LLM, while other work incorporates expert commentary on source material to help models learn relationships between examples and task objectives [88]. These approaches show that feedback can support alignment at multiple stages of the HAI pipeline, including before, during, and after training. For HAI-JDM, the central challenge is to **design feedback mechanisms that are scalable, representative of human goals, robust to bias, and useful for improving team-level decision-making rather than only optimizing isolated model behavior**.

4 Control and Decision Authority in Human-AI Teams

A central design question in human-AI joint decision-making is how decision authority should be allocated within the team. This section examines computational frameworks for shared control in HAI systems and asks a fundamental question: **who decides what, when, and why?** We show that control allocation is not merely an implementation detail, but a design choice that reflects assumptions about human capability, AI reliability, communication, trust, and alignment.

Existing approaches differ along two key dimensions. First, decision authority may be statically assigned, transferred between human and AI agents, or jointly negotiated through interaction. Second, humans may be modeled as fixed decision-makers whose capabilities and preferences are assumed in advance, or as adaptive collaborators whose expertise, trust, workload, and goals change over time. To clarify these assumptions, we organize prior work into three control allocation paradigms. In **fixed control allocation** (Section 4.1), authority is assigned in advance to either the human or the AI system. In **transfer based control allocation** (Section 4.2), authority can shift between agents through handoffs based on factors such as confidence, uncertainty, or expected utility. In **collaborative control allocation** (Section 4.3), authority is dynamically negotiated through interaction, allowing human and AI agents to adapt their roles as the decision process unfolds. These paradigms are illustrated in Figure 5.

4.1 Fixed Control Allocation: Authority is Assigned in Advance

Fixed control allocation assigns decision authority before the interaction begins. This paradigm is closely related to classical levels-of-automation frameworks, which describe how functions may be allocated between humans and automated systems across stages such as information acquisition, analysis, decision selection, and action implementation [111, 128]. In many HAI decision-support systems, the human retains final authority while the AI system provides recommendations, predictions, explanations, or alerts. In other systems, automation may retain primary control while the human monitors and intervenes when necessary. In both cases, the distribution of authority is largely predefined rather than negotiated during the decision process.

Fixed control allocation is common because it is simple to implement, easy to evaluate, and compatible with many human-in-the-loop and supervisory-control systems. However, prior work on automation and adaptive interfaces shows that static allocations can be brittle when task demands, human workload, situation awareness, or automation reliability change over time [66, 74]. If the human always retains control, the system may fail to exploit AI capabilities in regions where the model is more reliable. If the AI always retains control, the system may underuse human expertise, contextual judgment, or accountability. This motivates transfer-based approaches, which allocate decision authority at the instance level rather than fixing it in advance.

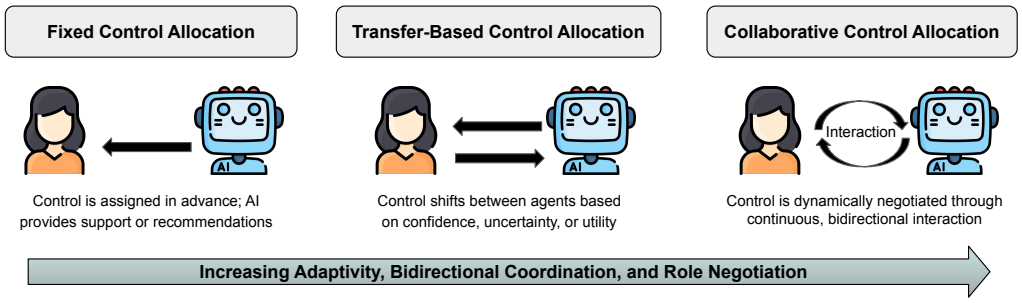


Fig. 5. Three paradigms of control allocation in human-AI decision-making. In fixed control allocation, decision authority is assigned in advance, with one agent leading and the other providing support or recommendations. In transfer-based control allocation, authority shifts between human and AI agents through discrete handoffs, often based on confidence, uncertainty, risk, or expected utility. In collaborative control allocation, authority is dynamically negotiated through continuous, bidirectional interaction, allowing both agents to adapt roles and coordinate decisions as the task unfolds.

4.2 Transfer-Based Control Allocation: Authority Shifts Through Handoffs

Transfer-based control allocation shifts decision authority between the human and the AI system through discrete handoffs. Rather than assigning authority once in advance, these approaches ask whether a particular instance should be handled by the model or by a human decision-maker. This paradigm has roots in classical machine learning work on classification with a rejection option. In this setting, a classifier abstains when its confidence is too low, thereby avoiding predictions that are likely to be incorrect. Chow [32] formalized this intuition through a Bayesian rejection rule in which the classifier accepts a prediction only when the maximum posterior probability exceeds a threshold. Specifically, for an input pattern v, the classifier computes the maximum posterior probability $m(v) = \max_i P(i \mid v)$, and accepts the prediction only when this confidence exceeds a threshold determined by the cost of rejection: Accept v if $m(v) \geq 1 - t$, Reject v if $m(v) < 1 - t$, where $t \in [0, 1]$ represents the rejection-cost parameter. This formulation allocates decision authority based purely on the model's posterior confidence: the model acts when confidence is sufficiently high and abstains otherwise.

Later work extended abstention through confidence-based rejection and cost-sensitive decision frameworks [13, 34]. These methods are typically evaluated using accuracy under a reject option, expected loss, or risk-coverage tradeoffs, which measure how well a model avoids low-confidence predictions. However, classical rejection approaches do not fully model collaboration. Rejection is treated as abstention from prediction, often assuming that an external oracle will resolve the case. This assumption is limited in HAI settings because human decision-makers are not error-free and may vary in expertise, availability, and access to contextual information.

4.2.1 Control Is Transferred Based on Confidence Through Deferral. The transition from rejection learning to **learning to defer** represents a shift from abstention to explicit human-AI decision routing. In learning to defer, the system learns not only a prediction, but also whether to make the decision itself or defer to a human decision-maker. Madras et al. [100] introduced an early formulation of this problem, where the model observes input features, produces a prediction, and also produces a binary deferral decision indicating whether authority should be transferred to the human. This framing treats collaboration as conditional delegation. The model acts when it is expected to perform well and defers when the human is expected to be more reliable.

Mozannar and Sontag [108] further developed this view through a decision-theoretic analysis of optimal deferral. Their framework compares the expected loss of model prediction with the expected loss of expert prediction, allowing

the system to defer when the expert is more likely to be correct. Formally, the Bayes-optimal deferral rule can be written as: $r^B(x) = \mathbf{1}\left[\max_{y\in\mathcal{Y}} \eta_y(x) \leq P(Y = M \mid X = x)\right]$, where $\eta_y(x) = P(Y = y \mid X = x)$ denotes the model posterior for class y, and $P(Y = M \mid X = x)$ denotes the probability that the expert prediction M is correct for instance x. Under this rule, the system defers when the expert is estimated to be more reliable than the model's most confident prediction. Compared with classical rejection learning, this formulation replaces a fixed confidence threshold with an instance-specific comparison between model and human competence. Learning-to-defer systems are commonly evaluated using expected loss, system-level accuracy, or joint human-AI performance, which capture whether routing decisions improve the final outcome relative to either agent acting alone.

However, deferral frameworks rely on several assumptions that may not hold in realistic HAI settings. They often require stable human decision labels during training, assume that human and model errors can be estimated reliably, and may ignore unobserved contextual information that humans use when making decisions. Subsequent work has relaxed these assumptions in several ways. Hemmer et al. [61] model human expertise to support deferral with limited human labels. Gao and Yin [46] address settings where human decisions depend on unobserved information. Joshi et al. [73] extend deferral to sequential decision problems, where deferring early can improve downstream outcomes. In online and resource-constrained settings, Gao et al. [47] and Reid et al. [120] model deferral under limited, delayed, or costly human feedback. These extensions improve the robustness of transfer-based control under more realistic data constraints. Yet deferral remains fundamentally a handoff mechanism, where authority is transferred to either the model or the human, rather than jointly exercised through interaction.

4.2.2 Control Is Transferred Based on Relative Competence Through Complementarity. **Learning to complement** shifts the focus from model confidence to relative competence. Instead of asking only when the model should defer, complementarity-based approaches ask where the model should specialize so that the combined human-AI system performs better than either agent alone (Table 1). Wilder et al. [160] formulate this problem by training models to perform well in regions of the input space where humans are likely to make errors. This represents a conceptual shift from reactive delegation to proactive role specialization. This idea can be expressed as an objective that trades off model-only utility against the utility of incorporating human input. A common way to formalize this tradeoff is $\mathbb{E}_{(x,y,h)\sim P}\left[q_\theta(x)\left(u(y, m_\theta(x, h)) - c\right) + (1 - q_\theta(x))u(y, m_\theta(x))\right]$. Here, x is the input, y is the ground-truth label, and h is the human decision. The function $m_\theta(x)$ denotes the model's independent decision, while $m_\theta(x, h)$ denotes the model's decision when incorporating human input. The query policy $q_\theta(x)$ determines whether to solicit human input, $u(\cdot)$ measures task utility, and c represents the cost of querying the human. This objective captures the core idea of complementarity: human input should be used when it improves team utility enough to justify its cost.

Complementarity approaches are typically evaluated using team utility, joint loss, or system-level performance relative to human-alone and AI-alone baselines. These metrics are important because complementarity is not captured by model accuracy alone. A highly accurate model may add little value if it succeeds and fails on the same cases as the human, while a less accurate model may improve team performance if its errors are weakly correlated with human errors. Steyvers et al. [137] emphasize this point by showing that hybrid performance depends strongly on error structure, not simply on the individual accuracy of the human or AI system.

Subsequent work extends complementarity to more realistic collaboration settings. Some approaches incorporate noisy multi-rater human input [168], while others unify deferral and complementarity within a single decision space [169]. These extensions move beyond single-expert, clean-label assumptions toward multi-expert, noise-aware, and cost-sensitive collaboration. However, they still largely preserve a transfer-based view of control. Decision authority is

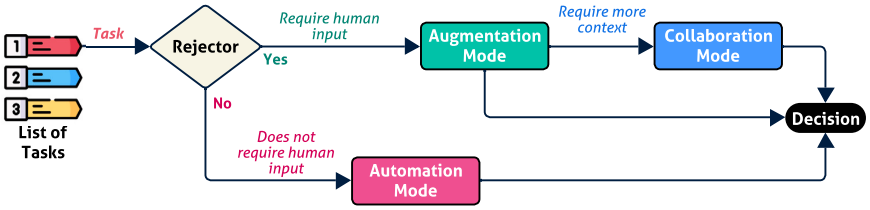


Fig. 6. The A^{2}C decision flow from Tariq et al. [143], with permission. The framework distinguishes between accepting an AI decision, assisting a human decision-maker, and entering a collaborative mode where human and AI agents iteratively exchange information under uncertainty.

allocated to either the human or the AI on a per-instance basis. As a result, these systems can improve team performance through better routing and specialization, but they do not fully capture settings in which humans and AI systems iteratively communicate, revise beliefs, or negotiate authority during the decision process.

4.3 Collaborative Control Allocation: Authority is Negotiated Through Interaction

Transfer-based control allocation treats authority as a routing problem where, for each instance, either the human or the AI system should decide. This framing is useful when uncertainty can be resolved by selecting the more competent agent. However, it is limited in settings where both human and AI agents are uncertain, where their information is partial or complementary, or where the decision requires iterative refinement. In such cases, the goal is not simply to transfer authority, but to share and negotiate it through interaction.

The motivation for collaborative control is consistent with research on shared leadership in human teams. Shared leadership, in which multiple team members influence the team's direction and decision process, has been associated with higher satisfaction, problem-solving quality, objective team performance, trust, cohesion, and reduced conflict [4, 14, 19, 105, 113]. The implication for HAI is not that AI systems should simply become team leaders, but that decision authority should be flexible enough to reflect changing expertise, uncertainty, workload, and task context. Recent HAI work begins to operationalize this idea by treating collaboration as a distinct control mode. Tariq et al. [143] propose the A^2C framework, which introduces three modes: *accept*, *assist*, and *collaborate*: $m(x) \in \{\text{accept}, \text{assist}, \text{collaborate}\}$, where $m(x)$ denotes the control mode selected for instance x. In the *accept* mode, the classifier makes the decision. In the *assist* mode, the human expert makes the final decision with support from model outputs, confidence information, or explanations. In the *collaborate* mode, uncertainty triggers iterative exchange between the human and AI system, allowing both agents to update beliefs and refine the final decision. This method is illustrated in Figure 6.

The A^{2}C framework is important because it treats collaboration as more than abstention or deferral. Rather than viewing uncertainty only as a reason to hand off control, it frames uncertainty as a reason for joint inference. Empirically, the framework is evaluated through task accuracy and decision quality, showing that collaborative modes can improve joint performance, particularly as human competence increases. This makes A^{2}C a useful example of explanation-mediated interaction as a mechanism for shared control. At the same time, A^{2}C still activates collaboration primarily in uncertain cases, rather than treating interaction as a continuous feature of the entire decision process. Thus, it moves beyond fixed and transfer-based paradigms, but remains a conditionally triggered collaboration architecture. More general collaborative control systems would allow human and AI agents to continuously negotiate roles, communicate uncertainty, update beliefs, and revise control authority as the task unfolds.

Table 1. Learning to defer and learning to complement approaches to transfer-based control allocation.

Dimension	Learning to Defer	Learning to Complement
Decision question	Should the AI decide, or should it defer to a human?	Which teammate, human or AI, is expected to perform better?
Human role	Fallback expert when the AI should abstain.	Complementary decision-maker with strengths that may differ from the AI's.
Training signal	Expert labels, model predictions, confidence, and/or deferral cost.	Logged human responses, model predictions, and human–AI complementarity.
Objective	Minimize system loss by balancing prediction error and deferral cost.	Maximize expected team performance or team utility.

4.4 Synthesis: From Handoffs to Adaptive Role Negotiation

Across these paradigms, control allocation evolves from predefined authority, to instance-level handoffs, to interactive role negotiation. Fixed control allocation is simple and interpretable, but cannot adapt to changing human or AI capabilities. Transfer-based control improves flexibility by routing decisions based on confidence, competence, uncertainty, or expected utility, but still treats authority as something assigned to one agent at a time. Collaborative control moves closer to human-inspired teaming by allowing both agents to participate in the decision process, exchange information, and adapt roles over time. This progression also changes how HAI systems should be evaluated. Fixed and transfer-based systems are often evaluated through task accuracy, expected loss, risk-coverage tradeoffs, or joint utility. Collaborative control requires broader metrics, including communication quality, reliance calibration, workload, decision quality, recovery from errors, and longitudinal team performance. For HAI-JDM, the central challenge is to design control mechanisms that are not only accurate at routing decisions, but also capable of supporting adaptive, bidirectional, and context-sensitive teamwork.

5 Evaluating Human–AI Team Performance

Evaluation is central to HAI joint decision-making because it determines what kinds of systems are considered successful and drives the evolution of novel approaches. Much existing work evaluates human–AI systems through task accuracy or predictive performance [31, 163, 165]. These metrics are important, but they are insufficient for evaluating HAI teams because they do not capture whether humans and AI systems coordinate effectively, rely on one another appropriately, communicate uncertainty, or adapt over time. In HAI-JDM, the relevant unit of evaluation is not the AI system in isolation, nor the human user alone, but the joint human–AI team operating under an actual interaction policy.

We use **team utility** to refer to the value produced by the human–AI team as a collective decision-making unit. Team utility includes task outcomes, but is not reducible to accuracy. It may also include the cost of human effort, the appropriateness of reliance, the quality of communication, the allocation of decision authority, the human's mental model of the AI, and the team's ability to improve across repeated interactions. This section reviews how evaluation has been operationalized in the surveyed literature and argues that **HAI-JDM benchmarks should combine outcome, human-centered, interaction, control-allocation, and longitudinal team-level evaluation metrics**.

Existing HAI benchmarks and testbeds illustrate both the value and limitations of current evaluation practice. Controlled AI-advised decision-making testbeds such as CAJA evaluate how humans form mental models of AI systems and how those mental models affect reliance, acceptance of AI recommendations, and team performance [6]. Cooperative game benchmarks such as Overcooked-AI measure coordination through task reward, completed deliveries, efficiency,

Table 2. Evaluation metric categories for human–AI joint decision-making, synthesized from the surveyed literature.

Metric category	Examples from surveyed work	What it measures	Strengths	Limitations	Representative work
Task outcome and model performance	Accuracy, game score, completed deliveries, success rate, task reward, policy performance, sample efficiency, latency, decision quality, simulated safety	Whether the AI system or human–AI system produces correct, efficient, or safe task outcomes	Simple to compute; widely comparable; useful for measuring final task success	Often ignores how outcomes were produced; high model accuracy may not improve joint outcomes	[9, 25, 31, 139]
Human-centered and cognitive metrics	Trust, confidence, perceived understanding, perceived humanness, perceived usefulness, human preference, willingness to collaborate, mental model accuracy, belief calibration	How humans perceive, understand, trust, and mentally model the AI system	Captures cognitive and perceptual factors that task accuracy misses	Can be subjective and context-dependent; may be weakly tied to actual team performance	[8, 15, 20, 78]
Interaction and coordination metrics	Reliance behavior, acceptance/rejection of AI recommendations, overreliance, underreliance, reliance calibration, behavioral coordination, communication efficiency, uncertainty exchange, feedback timing	How humans and AI systems coordinate during the decision process	Evaluates the interaction process rather than only final outcomes	Observable behavior may not reveal why users relied on or rejected AI advice; coordination gains may not translate to better outcomes	[8, 91, 145, 148]
Control allocation and delegation metrics	Deferral rate, rejection/abstention, risk–coverage tradeoff, expected loss, query cost, handoff decisions, expert correctness, complementarity, joint loss	How decision authority is assigned, transferred, or negotiated between human and AI agents	Captures when the system should act, defer, request input, or specialize	Often assumes static human expertise; commonly models control as discrete handoff rather than continuous negotiation	[32, 108, 143, 160]
Team-level utility and longitudinal performance	Human–AI team accuracy, joint reward, collaborative reward, improvement over human-alone, AI-alone, or human–human baselines, team utility, recovery after errors, adaptation over repeated interactions	Whether the human–AI team produces value as a collective decision-making unit over time	Best aligned with HAI-JDM; can integrate outcomes, costs, reliance, coordination, and adaptation	Hard to standardize across tasks; may require longitudinal studies and explicit tradeoff modeling	[6, 7, 136, 151]

and adaptation to human partners [25], while the Hanabi Learning Environment evaluates cooperative reasoning under partial observability through game score, communication conventions, and theory of mind-like inference [12]. Human-autonomy testbeds such as CHAOPT further use Overcooked-style environments to study real-time team coordination, task performance, and human-autonomy collaboration [17]. These benchmarks are valuable because they operationalize different parts of HAI performance, but they also demonstrate the fragmentation of current evaluation. Some emphasize final task outcomes, others emphasize human mental models and reliance, and others emphasize communication, coordination, or adaptation. HAI-JDM therefore requires evaluation practices that combine these perspectives rather than optimizing a single benchmark score.

To organize this fragmented evaluation landscape, Table 2 summarizes the main metric categories that appear across the surveyed literature. The table distinguishes between metrics that evaluate final outcomes, human perceptions and cognition, interaction and coordination processes, control allocation, and team-level utility over time. This categorization highlights a central theme of this section: **each metric category captures an important part of HAI performance, but no single category is sufficient for evaluating joint decision-making as an adaptive team process.**

5.1 Why Accuracy Is an Insufficient Objective

A broad line of work shows that optimizing predictive accuracy does not reliably translate into improved HAI team performance [7, 44, 136]. Accuracy evaluates whether a model or team produces the correct answer, but it does not

reveal whether the human and AI system coordinated effectively, whether the human relied on the AI appropriately, or whether the AI system contributed value beyond what the human could achieve alone.

Bansal et al. [7] formally show that log-loss-optimal, highly accurate models can be suboptimal under realistic human decision policies, and that training explicitly for team utility can produce better joint outcomes than simply improving model accuracy. Fragiadakis et al. [44] and Steyvers and Kumar [136] similarly argue that evaluation must move beyond individual accuracy to include joint performance and the human factors that shape it. In particular, Steyvers and Kumar [136] emphasize that accurate mental models of the system and appropriate interaction structures are necessary for effective reliance and improved team utility. Vaccaro et al. [151] further shows that HAI teams can underperform solo decision-makers in judgment-heavy tasks, even when paired with a highly accurate AI teammate.

These findings motivate a shift from evaluating AI systems as standalone predictors to evaluating HAI systems as interactive decision-making teams. Accuracy remains necessary, but it is not sufficient. A model can be accurate yet poorly used, trusted for the wrong reasons, ignored when useful, or harmful when its errors are systematically accepted. Conversely, a less accurate model may still improve team utility if it complements human expertise, communicates uncertainty effectively, and supports calibrated reliance.

5.2 Operationalizing Team Utility

Team utility can be operationalized in different ways depending on the task, interaction structure, and role of the AI system. Across the surveyed literature, we identify several recurring approaches: measuring joint outcomes, measuring reliance and calibration, measuring human reasoning and mental models, and measuring control allocation or complementarity. These approaches differ in what part of the HAI decision-making process they capture.

5.2.1 Joint outcomes capture team performance but not the process that produced it. The most direct way to measure team utility is to compare the performance of the human–AI team against relevant baselines, such as human-alone, AI-alone, and, where appropriate, human–human performance. This approach appears in collaborative game environments, diagnostic decision support, autonomous driving simulations, and control-allocation frameworks, where metrics include team score, completed deliveries, diagnostic accuracy, success rate, joint reward, or collaborative reward. Such metrics are useful because they evaluate whether the team actually produces better decisions. However, joint outcomes alone do not explain why the team succeeds or fails. High joint accuracy may reflect appropriate coordination, but it may also reflect one agent dominating the decision process. Poor joint performance may result from low AI quality, poor human reliance, misleading explanations, bad timing, or ineffective authority allocation. As a result, joint outcome metrics should be interpreted alongside measures of interaction, reliance, communication, and control.

5.2.2 Explanation and confidence signals shape reliance but do not guarantee better team outcomes. A growing line of work examines how confidence signals and explanations shape human reliance on AI-assisted decision-making. Bansal et al. [8] show that explanations can increase acceptance of AI recommendations regardless of correctness, rather than reliably improving team utility in complementary teams. This suggests that transparency can influence reliance behavior without ensuring that such reliance is appropriate. Similarly, Steyvers et al. [138] show that users may systematically overestimate model reliability when explanations are longer, even when explanation length does not correlate with correctness. These findings show that explanation and confidence metrics should not be treated as direct proxies for team utility. Explanation usefulness, perceived understanding, or confidence may improve while joint performance remains unchanged or worsens. Evaluation should therefore ask not only whether an explanation is

understandable or persuasive, but whether it improves calibrated reliance, helps humans identify when the AI is likely to be wrong, and leads to better joint decisions.

5.2.3 Human reasoning and mental models shape team utility beyond observable outcomes. Team utility also depends on the quality of human reasoning during interaction. Buçinca et al. [20] show that overreliance can occur when users accept AI recommendations without sufficient scrutiny. By introducing cognitive forcing functions that prompt users to reflect more carefully on AI outputs, they reduce unsafe reliance and improve joint accuracy. Similarly, Chiang et al. [29] show that AI-generated counterarguments can improve team decision-making by increasing deliberation depth and critical evaluation. These studies expand evaluation beyond final accuracy by treating reasoning engagement, deliberation quality, and reliance calibration as part of team utility. This is important because HAI decision-making is not only a matter of selecting the right output, but also of supporting the cognitive processes through which humans evaluate, contest, or integrate AI contributions. However, reasoning-focused metrics can be difficult to standardize because they are often task-specific and may require qualitative coding, behavioral traces, or process-level measurements.

5.2.4 Cognitive benchmarks are useful but incomplete proxies for collaborative capability. Recent benchmarks evaluate whether AI systems exhibit cognitive capabilities relevant to collaboration, such as Theory of Mind, mental modeling, and trustworthiness [41, 71, 92, 93, 129, 130, 155]. These benchmarks are relevant to HAI-JDM because collaboration depends on whether AI systems can reason about human beliefs, preferences, reliability, and social context. Systems with stronger mental-modeling or trustworthiness scores may be better positioned to support coordination, especially in tasks requiring perspective-taking or adaptation. However, cognitive benchmark performance should not be equated with team utility. Models may perform well on Theory of Mind or trustworthiness benchmarks without demonstrating robust collaborative behavior in real interactions [95]. Similarly, models can sometimes be optimized to score higher on trustworthiness benchmarks without becoming more useful or reliable teammates [68]. These limitations suggest that cognitive benchmarks should be used as diagnostic tools, not as substitutes for human-subject evaluation or team-level performance measures.

5.2.5 Control-allocation metrics evaluate whether authority is routed effectively. As discussed in section 4, deferral, rejection, complementarity, and collaborative control frameworks evaluate team utility by asking when the AI should act, when it should defer, and when human input should be solicited. Metrics such as risk–coverage tradeoffs, expected loss, query cost, system-level accuracy, complementarity, and joint loss capture whether decision authority is allocated effectively across human and AI agents. These metrics are useful because they explicitly evaluate the value of combining human and AI capabilities. However, many such frameworks still model control as a single-shot routing problem and assume relatively stable human expertise. For HAI-JDM, future evaluations should extend control-allocation metrics to settings where authority is negotiated over time, human trust and workload change, and both human and AI agents update their beliefs through interaction.

5.3 Trade-offs and Failure Modes

Across these metric categories, we identify three recurring failure modes. First, **outcome-based metrics** such as individual accuracy, joint accuracy, task reward, or success rate evaluate the final outcome but are agnostic to the interaction process that produced it. As a result, they cannot distinguish effective collaboration from overreliance, underuse of human expertise, or one agent dominating the decision process. A team may achieve high performance while still exhibiting poor communication, miscalibrated trust, or brittle authority allocation.

Second, **interaction metrics** such as reliance calibration, acceptance or rejection of AI recommendations, deferral behavior, or communication frequency capture observable human responses to AI outputs but do not necessarily explain the causes of those behaviors. Explanations, confidence scores, or rationales may shift user behavior without improving decision quality. In these cases, interaction metrics may suggest improved engagement or reliance while true team utility remains unchanged or declines.

Third, **human-centered and cognitive metrics** such as trust, confidence, perceived understanding, mental model accuracy, and reasoning engagement provide insight into the psychological mechanisms that shape collaboration, but they trade off consistency and generalizability. They are often subjective, task-dependent, and difficult to compare across studies. They can also diverge from performance: users may feel more confident or perceive an AI system as more understandable even when their reliance is poorly calibrated or their decisions do not improve.

These failure modes show that no single metric category is sufficient for evaluating HAI-JDM. Outcome metrics capture what decision was made, interaction metrics capture how humans and AI systems responded to one another, cognitive metrics capture how humans understood or experienced the collaboration, and control-allocation metrics capture how authority was distributed. Effective evaluation requires connecting these pieces rather than optimizing them independently.

5.4 Implications for Evaluation Benchmarks

These trade-offs suggest that future HAI-JDM benchmarks should be multi-dimensional, comparative, and longitudinal. First, benchmarks should report **comparative baselines**: human-alone performance, AI-alone performance, human–AI team performance, and, when relevant, human–human team performance. Without these baselines, it is difficult to determine whether the AI system adds collaborative value or simply performs well in isolation.

Second, benchmarks should combine **outcome metrics with process metrics**. Reporting accuracy or task reward alone is insufficient; evaluations should also measure reliance behavior, acceptance and rejection of AI recommendations, communication quality, feedback use, deferral decisions, and recovery after errors. These measures help distinguish teams that perform well because of effective coordination from teams that perform well despite poor interaction dynamics.

Third, benchmarks should include **human-centered and cognitive measures** that assess whether humans form accurate mental models of the AI system, calibrate trust appropriately, and engage in effective reasoning. However, these measures should be paired with behavioral and outcome measures rather than treated as standalone indicators of success. Perceived understanding, trust, or satisfaction is valuable only insofar as it supports appropriate reliance and better joint decision-making.

Finally, HAI-JDM benchmarks should evaluate **adaptation over time**. Human beliefs about AI capability, trust, reliance, and expectations evolve through interaction, while AI systems may adapt to human feedback, preferences, or behavior. Static one-shot evaluations cannot capture these dynamics. Future benchmarks should therefore measure whether human–AI teams improve across repeated interactions, whether reliance becomes better calibrated, whether control allocation adapts to changing uncertainty or workload, and whether the team becomes more robust after errors.

In summary, HAI-JDM evaluation should not ask only whether the AI system is accurate or whether the human user is satisfied. It should ask whether the human–AI team makes better decisions together, whether the interaction process supports calibrated and adaptive collaboration, and whether the team improves over time. Team utility provides a useful organizing objective for this broader evaluation agenda, but it should be operationalized as a family of task-specific, interaction-aware, and longitudinal measures rather than as a single universal score.

6 Toward Human–AI Joint decision-making

The prior sections show that HAI decision-making research has made substantial progress on individual mechanisms of collaboration. Modeling humans allows AI systems to reason about human beliefs, goals, bounded rationality, and trust. Communication and feedback mechanisms support coordination, explanation, alignment, and preference learning. Control-allocation methods determine when authority should remain fixed, transferred between agents, or shared through interaction. Finally, evaluation work shows that accuracy alone is insufficient, and that HAI systems must be assessed through team utility, reliance calibration, interaction quality, and adaptation over time.

However, these research threads are often studied in isolation. Existing work still frequently treats HAI decision-making as an AI-assisted prediction problem, a one-shot recommendation task, or a routing problem in which authority is assigned to either the human or the AI. These framings are useful for lower-coordination tasks, but they do not fully capture settings in which humans and AI systems must coordinate repeatedly, update beliefs, communicate uncertainty, negotiate authority, and jointly adapt as the decision process unfolds.

To synthesize the surveyed literature, we first introduce a task taxonomy for HAI-JDM that clarifies how coordination requirements increase across task types. We then use this taxonomy to motivate the HAI-JDM framework, which organizes human and agent modeling, communication and alignment, control sharing, and team-level evaluation into a repeated adaptive decision cycle.

6.1 A Task Taxonomy for Human–AI Joint decision-making

Not all HAI decision-making tasks require the same degree of collaboration. Some tasks can be handled through one-way assistance or predefined role allocation, while others require continuous coordination, flexible role assignment, and shared control between human and AI agents. Drawing on human team task interdependence [123], we organize HAI-JDM tasks by increasing sub-task interdependence and coordination demand, as shown in Table 3.

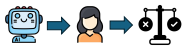
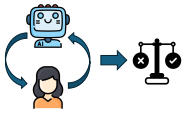
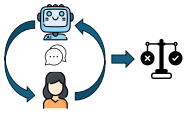
This taxonomy distinguishes five task types. **Ensemble-style pooled tasks** require little coordination because human and AI outputs are aggregated after being produced independently. **Routing-style pooled tasks** require predefined role allocation, where a mechanism assigns each instance to either the human or AI system. **Stepwise tasks** require temporal coordination because outputs from one stage affect later stages in a predefined pipeline. **Static team tasks** require information sharing and role definition during the task, but roles remain relatively stable. Finally, **dynamic team tasks** require high coordination because roles, authority, and sub-task responsibilities may change as the task unfolds.

This taxonomy clarifies a central gap in the surveyed literature. Many HAI decision-making frameworks remain concentrated in lower- or mid-coordination settings, where roles are predefined, interaction is limited, or control is transferred through discrete handoffs. These settings are important, but they do not fully capture dynamic team tasks, where human and AI agents must jointly determine what to do, who should do it, when authority should shift, and how the team should adapt based on feedback. Addressing dynamic team tasks requires integrating the mechanisms reviewed throughout this survey: mutual modeling, bidirectional communication and alignment, adaptive control sharing, and team-utility-based evaluation.

6.2 From Low-Coordination Assistance to Dynamic Teaming

The taxonomy illustrates why existing HAI approaches are not sufficient for the full range of joint decision-making tasks. In ensemble-style and routing-style pooled tasks, humans and AI systems can contribute without extensive

Table 3. Taxonomy of Human–AI Joint decision-making (HAI-JDM) task types, adapted from Saavedra et al. [123]. Task types are ordered by increasing sub-task interdependence and coordination demand. Existing HAI decision-making frameworks often address pooled, routing, stepwise, or static team settings, while dynamic team tasks remain comparatively underdeveloped.

Task type	Key requirement	Description	Representative HAI application	Visual
Ensemble-style pooled	Output aggregation with little or no prior coordination	**Low coordination:** Human and AI outputs are produced independently and aggregated into a single decision.	Human–AI ensembles combine human and model outputs without intra-team communication [30].	
Routing-style pooled	Predefined role allocation	**Low coordination:** A routing mechanism assigns each input to either the human or AI system.	Learning to Defer and Learning to Complement allocate decisions based on predicted human or AI competence [100, 160].	
Stepwise	Temporal coordination across predefined stages	**Moderate coordination:** Subtasks occur in a fixed order, and outputs from earlier stages influence later stages.	RLHF uses human evaluations of AI outputs to update model behavior within a predefined training pipeline [33, 77, 172].	
Static team	Information sharing with relatively stable roles	**Moderate coordination:** Team members communicate during the task, but roles and authority remain mostly predefined.	Explanatory Interactive Learning combines AI predictions and explanations with human corrective feedback [144].	
Dynamic team	Flexible roles and adaptive control allocation	**High coordination:** Team members jointly determine subtasks, roles, and authority as the decision process unfolds.	Mixed-initiative human–robot teaming supports adaptive task allocation between humans and AI agents with asymmetric capabilities [109].	

interaction. Their outputs are aggregated, or a routing mechanism assigns authority to one agent. In stepwise tasks, interaction becomes more structured over time, but the sequence of roles is still largely predefined. In static team tasks, communication and feedback become more important, but role structure remains relatively stable. Dynamic team tasks require a stronger form of collaboration. In these settings, the human and AI system cannot rely only on predefined roles, one-shot explanations, or fixed deferral policies. Instead, they must continuously model one another, communicate uncertainty and intent, align goals, and negotiate control as the task unfolds. This is the point at which HAI systems must move from assistance or delegation toward teaming.

The surveyed literature provides many of the components needed for this shift, but rarely integrates them into a single adaptive process. Human modeling work often focuses on predicting human actions or preferences. Communication work often focuses on explanations, feedback, or alignment mechanisms. Control-allocation work often focuses on routing authority. Evaluation work increasingly recognizes the importance of team utility. The challenge is to connect these mechanisms so that they support dynamic, interdependent decision-making rather than isolated improvements to individual parts of the system.

6.3 The HAI-JDM Framework

Based on this synthesis, we propose HAI-JDM as a unifying conceptual framework for studying, designing, and evaluating HAI joint decision-making systems (Figure 7). The framework is intended especially for higher-coordination

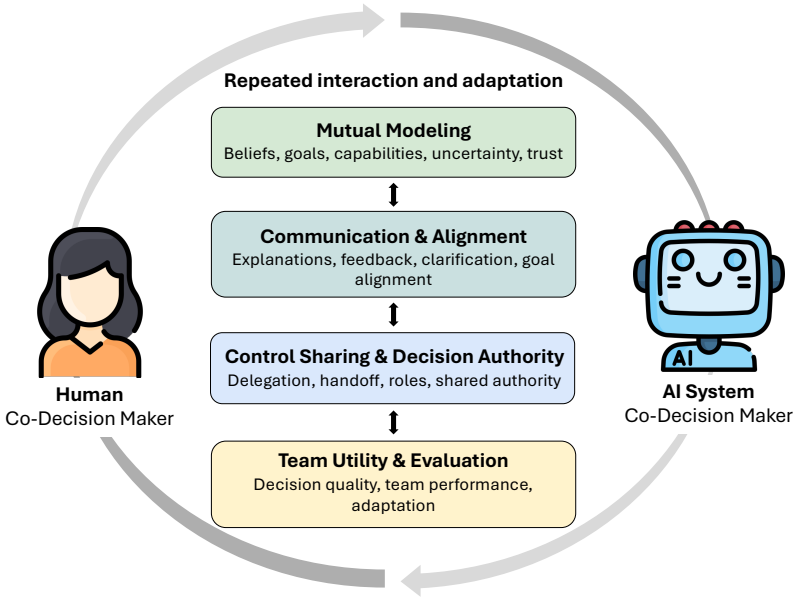


Fig. 7. HAI-JDM framework. Humans and AI systems are treated as co-decision-makers who coordinate during joint decision formation through human and agent modeling, communication and alignment, and control sharing. Decision outcomes are evaluated using team utility, which feeds back to update modeling, communication, and control in subsequent interactions, enabling adaptation over repeated joint decision cycles.

tasks, where humans and AI systems act as interdependent co-decision-makers rather than as isolated predictors, passive users, or static experts. Under HAI-JDM, humans and AI systems coordinate during joint decision formation through three core mechanisms: human and agent modeling, communication and alignment, and control sharing. Decision outcomes are then evaluated through team utility, which feeds back into subsequent interactions and enables adaptation over time.

The framework has four components. First, **human and agent modeling** captures how each collaborator represents the other's goals, beliefs, capabilities, limitations, uncertainty, and likely behavior. This includes AI models of human cognition, bounded rationality, trust, and Theory of Mind, as well as human mental models of AI reliability and competence. Second, **communication and alignment** captures how humans and AI systems exchange information, communicate uncertainty, provide feedback, and align goals and strategies. Third, **control sharing** captures how decision authority is assigned, transferred, or negotiated throughout the decision process. Fourth, **team utility** evaluates whether the human–AI team produces value as a collective decision-making unit.

To make these relationships more precise, we introduce a lightweight notation for the adaptive interaction cycle captured by HAI-JDM. This notation is not intended to define a single algorithmic model for all HAI joint decision-making systems. Rather, it provides a compact way to describe the dependencies that recur across the literature reviewed in this survey. Let x_t denote the task state at decision cycle t, H_t denote the human state, including both observable behavior and latent factors such as goals, beliefs, trust, workload, expertise, preferences, and mental models of the AI system, and A_t denote the AI state. The human state may include goals, beliefs, expertise, workload, trust, preferences, and mental models of the AI system. The AI state may include its task model, uncertainty estimates, policy, learned representation of the human, and estimates of human goals or capabilities. At each decision cycle, the human and AI

system exchange communication and alignment signals C_t, determine a control-sharing arrangement α_t, and produce a joint decision or action d_t:

$$C_t = f_C(x_t, H_t, A_t), \tag{1}$$

$$\alpha_t = f_\alpha(x_t, H_t, A_t, C_t), \tag{2}$$

$$d_t = f_D(x_t, H_t, A_t, C_t, \alpha_t). \tag{3}$$

The resulting decision changes the task state and also updates the human and AI states through feedback, observation, and interaction history:

$$(x_{t+1}, H_{t+1}, A_{t+1}) = f_U(x_t, H_t, A_t, C_t, \alpha_t, d_t, o_t), \tag{4}$$

where o_t represents observed outcomes, feedback, or other evidence produced by the decision process. Team performance can then be represented abstractly as a cumulative team utility:

$$U_{\text{team}} = \sum_{t=0}^{T} \gamma^t u(x_t, H_t, A_t, C_t, \alpha_t, d_t, o_t), \tag{5}$$

where $u(\cdot)$ may encode task performance, decision quality, human-centered outcomes, coordination costs, fairness, safety, accountability, or other criteria relevant to the application context. In this notation, H_t and A_t correspond to mutual modeling, C_t to communication and alignment, α_t to control sharing, and U_{team} to team-level evaluation. This formulation emphasizes that HAI-JDM is not defined by any single reward function or optimization procedure. Instead, it represents HAI joint decision-making as an adaptive process in which modeling, communication, control sharing, and evaluation are mutually dependent over time.

A central feature of HAI-JDM is that team utility is not only an endpoint metric. It also functions as feedback for future adaptation. Poor team utility may indicate inaccurate human models, miscalibrated trust, insufficient communication, or inappropriate control allocation. Strong team utility may reveal useful complementarities, effective communication strategies, or successful role negotiation. Over repeated decision cycles, these signals should update how humans and AI systems model one another, communicate, and share control.

7 Open Challenges and Research Directions

7.1 Modeling Collaborators Dynamically and Bidirectionally

Dynamic HAI teams require collaborators to maintain evolving models of one another. As summarized in Table 4, current work has made progress on bounded rationality, Theory of Mind, and trust modeling, but these approaches often remain static, one-directional, or limited to narrow task settings. For HAI-JDM, modeling should support mutual adaptation. AI systems should update beliefs about human goals, expertise, uncertainty, rationality, and trust, while humans should be supported in forming accurate mental models of AI capabilities and limitations.

The central open challenge is to move from one-directional human modeling to mutual modeling. Existing systems often model humans as fixed sources of labels, preferences, demonstrations, or actions. Dynamic HAI teams instead require models that evolve over repeated interactions and influence communication and control decisions. For example, a system's estimate of human expertise or trust should affect when it explains, asks for clarification, defers, or takes initiative. Likewise, the system should communicate its own uncertainty and limitations so that humans can update their mental models of the AI over time.

Table 4. Open challenges and recommendations for modeling collaborators in HAI-JDM.

Current direction	Limitations	Recommendation
Bounded-rationality modeling. Human behavior can be modeled as approximately rational, as in Pragmatic Pedagogic Value Alignment [43].	Often assumes a fixed degree of rationality and is primarily developed in reinforcement-learning settings.	**Model bounded rationality dynamically.** Represent human decision-making as context-dependent and changing over time, rather than as perfectly rational or fixed.
Theory of Mind modeling. ToM-based and human-aware models can improve coordination and human preference for AI teammates [25, 27, 31, 166].	Often requires costly human interaction data and may not generalize across people, tasks, or strategies.	**Model beliefs, goals, and strategies.** AI teammates should infer human beliefs, objectives, uncertainty, and current strategy while remaining robust to human variability.
Trust modeling. Trust can be modeled as an evolving latent state shaped by interaction history and system behavior [90, 148, 164].	Most work focuses on human trust in AI, while AI models of human reliability and bidirectional trust remain underdeveloped.	**Model trust bidirectionally.** Track how trust, reliance, and perceived competence evolve for both collaborators and use these estimates to guide communication and control.

Table 5. Open challenges and recommendations for communication and alignment in HAI-JDM.

Current direction	Limitations	Recommendation
Implicit and explicit communication. AI systems increasingly use explanations, rationales, confidence signals, and implicit social or action-based cues to support coordination [8, 28, 39, 81, 91, 159].	Communication is often one-way, post-hoc, or limited to explicit explanations. Current systems still struggle to interpret and generate the nuanced cues humans use in teamwork.	**Support bidirectional communication of goals and uncertainty.** AI teammates should communicate goals, strategies, uncertainty, and intent, while also interpreting human feedback, corrections, and implicit coordination cues.
Feedback-based alignment. RLHF, expert-guided training, and preference-based methods use human input to align model behavior with human goals [9, 16, 33, 88].	Alignment is often treated as a training-stage objective or one-shot feedback process. Human feedback can be costly, sparse, biased, or difficult to update as goals and contexts change [86].	**Make alignment ongoing and actionable.** HAI systems should treat feedback as part of continuous team coordination, allowing both human and AI collaborators to revise goals, strategies, and behavior over time.

7.2 Communication and Alignment Beyond Explanation

Higher-interdependence tasks require communication that is continuous, bidirectional, and actionable. As summarized in Table 5, current work has made progress on implicit communication, explanations, feedback, RLHF, and expert-guided alignment. However, many approaches still treat communication as a one-way channel: the AI explains to the human, the human provides feedback to the AI, or preferences are collected before deployment. In dynamic HAI teams, communication should instead support ongoing goal alignment, uncertainty exchange, disagreement resolution, and strategy coordination.

The main research direction is to treat communication as a mechanism for team adaptation, not only as an interface feature. Explanations, confidence scores, implicit cues, and feedback should be evaluated by whether they improve shared understanding and team utility, not merely whether they increase user confidence or acceptance. This is especially important for static and dynamic team tasks, where subtasks are interdependent and communication failures can cascade into misaligned actions or inappropriate control allocation.

Table 6. Open challenges and recommendations for control sharing in HAI-JDM.

Control paradigm	Limitations	Recommendation
Fixed Control Allocation. Decision authority is assigned in advance, with one agent leading and the other providing support or supervision.	Useful for simple or well-scoped tasks, but brittle when human expertise, AI reliability, workload, trust, or task demands change over time.	**Move beyond static role assignment.** Use fixed control only when task demands and agent capabilities are stable; otherwise support mechanisms for revising authority during interaction.
Transfer-Based Control Allocation. Authority shifts through discrete handoffs, such as rejection, deferral, delegation, or complementarity-based routing [32, 34, 100, 108, 160].	Improves flexibility over fixed control, but often treats humans as static experts and models control as a one-shot routing decision rather than an evolving team process.	**Make handoffs context-aware and adaptive.** Update deferral and routing policies using human expertise, AI uncertainty, workload, trust, interaction history, and feedback from prior decisions.
Collaborative Control Allocation. Emerging approaches introduce collaborative or mixed-initiative modes in which humans and AI systems jointly participate in decisions [109, 143].	Moves toward teaming, but collaboration is often triggered only under uncertainty or constrained to predefined modes; continuous role negotiation remains underdeveloped.	**Support adaptive role negotiation.** Enable humans and AI systems to revise roles, share authority, request assistance, communicate uncertainty, and shift initiative as the decision process unfolds.

7.3 Control and Decision Authority as Adaptive Role Negotiation

Dynamic team tasks require more flexible control allocation than current deferral, routing, and complementarity methods provide. As summarized in Table 6, existing approaches often allocate authority through fixed policies, one-shot delegation, or instance-level routing. These methods are useful for routing-style pooled tasks, but they do not fully capture settings where human and AI roles must change as the task unfolds.

The key open challenge is to move from handoff-based control to interaction-driven control sharing. Future systems should update control policies using interaction history, task context, human state, AI uncertainty, and feedback from prior decisions. In HAI-JDM, control sharing is tightly connected to modeling and communication: the system cannot decide who should lead without modeling both collaborators, and control cannot be negotiated without mechanisms for communicating uncertainty, intent, and constraints.

7.4 Evaluation: From Accuracy to Team Utility

Finally, dynamic HAI teams require evaluation methods that go beyond one-shot accuracy or subjective satisfaction. Section 5 showed that existing metrics capture different fragments of the HAI decision process, such as task outcomes, human perceptions, interaction behavior, control allocation, and longitudinal team utility. The open challenge is to combine these metrics into benchmarks that reflect the requirements of higher-coordination tasks.

For HAI-JDM, evaluation should ask whether the human–AI team improves over repeated decision cycles, whether reliance and trust become better calibrated, whether communication supports shared strategy formation, and whether control allocation adapts to changing uncertainty and expertise. This also means that team utility should function as a feedback signal within the system, not only as an external evaluation measure. Poor team utility should trigger updates to human and agent models, communication strategies, and control-sharing policies. In this way, evaluation becomes part of the adaptive loop required for HAI-JDM.

These challenges show why HAI-JDM cannot be reduced to any single mechanism. Modeling without communication cannot support shared understanding; communication without adaptive control cannot support flexible role negotiation;

and control allocation without team-level evaluation cannot reveal whether the resulting collaboration is actually useful. The open research direction is therefore to design HAI systems in which modeling, communication, control, and evaluation are integrated into a single adaptive team process.

7.5 Agentic AI as an Emerging Direction

Agentic AI makes the need for HAI-JDM more urgent. As AI systems gain the ability to plan, use tools, maintain memory, and act over extended horizons, they may increasingly participate in decision-making as active collaborators rather than passive recommendation systems [58, 67, 118]. This creates opportunities for richer human–AI teaming, but it also intensifies the challenges identified by HAI-JDM. In particular, agentic systems may move HAI decision-making toward higher-coordination task types, where humans and AI systems must coordinate across multiple steps, revise goals, share context, and adapt to changing task conditions.

Agentic systems may support these higher-coordination tasks because they can participate in multi-step reasoning, maintain interaction histories, retrieve and integrate external information, and adapt behavior across time, including by using prior interactions as feedback signals for future adaptation [157]. However, autonomy alone does not guarantee effective collaboration. An agent may complete subtasks independently while still lacking accurate models of human goals, constraints, expertise, trust, or preferences. It may also fail to communicate uncertainty or intent, miscalibrate human reliance, or pursue objectives that are misaligned with team needs. Thus, agentic AI is not equivalent to HAI-JDM, as an autonomous system can still reproduce the limitations of earlier AI-assisted decision-making systems if it is not designed for mutual modeling, bidirectional alignment, adaptive control sharing, and team-utility-based evaluation.

Application domains illustrate both the promise and risk of this transition. In software engineering, education, scientific discovery, personal productivity, and robotics, agentic systems are increasingly positioned as collaborators that can decompose tasks, execute subtasks, monitor progress, and revise plans. These settings are natural candidates for HAI-JDM because the human and AI system may need to negotiate goals, divide work, communicate intermediate reasoning, and recover from errors over time. However, these same properties make evaluation more difficult. Success depends not only on whether the agent completes a task, but also on whether its actions are legible, interruptible, aligned with user intent, and responsive to human feedback [69, 106, 125].

Health and medicine provide a particularly important example of this transition. Many current LLM-based medical systems are still evaluated as decision-support tools, but their intended use increasingly points toward agentic workflows: monitoring longitudinal patient data, integrating context across time, generating personalized recommendations, asking clarifying questions, and supporting multi-step care decisions. For example, DM-Bench evaluates LLMs for personalized diabetes-management decision support using longitudinal continuous glucose monitoring and behavioral data, with metrics including accuracy, groundedness, safety, clarity, and actionability [23]. Although such benchmarks do not yet evaluate full agentic HAI teams, they illustrate why future medical HAI systems will require more than isolated answer accuracy. As medical decision-support systems become more interactive and agentic, they will need stronger mechanisms for human modeling, bidirectional alignment, adaptive control sharing, and team-utility-based evaluation.

Recent work on generative agents also highlights the potential for agentic AI to simulate social behavior and interaction patterns [112, 156]. These systems may provide useful testbeds for studying human-like coordination, but they also raise new risks. Agentic misalignment, in which AI agents pursue actions that diverge from human instructions or objectives, makes bidirectional alignment and controllability especially important [24, 98]. In HAI-JDM settings, this risk is not only about whether the agent follows a command, but whether it remains aligned with the evolving goals, constraints, and shared understanding of the human–AI team.

From the perspective of HAI-JDM, agentic AI should be evaluated not only by whether it completes tasks autonomously, but by whether it can function as a reliable member of a HAI decision-making team. This requires dynamic human and agent modeling, bidirectional communication and alignment, adaptive control sharing, and team-utility-based evaluation. Agentic AI is therefore a promising instantiation of HAI-JDM, but it is not sufficient on its own. Without explicit mechanisms for collaboration, agentic systems may scale up the same problems found in earlier AI-assisted decision-making systems: brittle human models, unidirectional communication, poorly calibrated trust, rigid control allocation, and evaluation that rewards task completion without measuring team quality.

8 Conclusion

As AI systems become increasingly involved in complex decision-making, HAI research must move beyond viewing AI as a passive tool for prediction, explanation, or recommendation and toward understanding humans and AI systems as adaptive decision-making teammates. This survey synthesized work on human modeling, communication and alignment, control allocation, and evaluation through the lens of human-inspired collaborative intelligence. Across these areas, we find that existing work has made substantial progress on individual mechanisms, but often studies them in isolation or in low-coordination settings where roles are predefined and interaction is limited. To address this gap, we introduced a task taxonomy for HAI joint decision-making and proposed HAI-JDM as a literature-grounded conceptual framework for understanding HAI-JDM as a repeated adaptive cycle of mutual modeling, bidirectional communication, control sharing, and team-utility-based evaluation. The central implication is that effective HAI teams cannot be achieved by optimizing accuracy, explanation quality, trust, or delegation independently; instead, future systems must integrate these mechanisms so that humans and AI systems can coordinate, revise beliefs, negotiate authority, recover from errors, and improve together over time. As agentic AI systems become more autonomous and embedded in high-stakes domains, this shift from AI-assisted decision-making to adaptive human–AI teaming will become increasingly important for building robust, accountable, and genuinely collaborative decision-making systems.

Acknowledgments

This work was supported by the National Science Foundation Graduate Research Fellowship Program under grant number 2234693. We used icons from FlatIcon by authors Vitaly Gorbachev and Freepik.

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